Jacob March

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Research Interests

Quantum information and quantum gravity: holographic entanglement entropy, reflected entropy and tensor networks, and the role of quantum error correction in holography. Also interested in computational and machine-learning methods applied to these questions, from decoding topological codes to the spectral statistics of SYK.

Education

Northeastern University, Boston, MA

Ph.D., Physics — Expected May 2029

M.A., Physics — May 2026

Grinnell College, Grinnell, IA

B.A., Mathematics & Physics, with Honors — May 2024

Publications

Refereed journal articles

[1] O. Gartlan, **J. March**, L. Rodriguez, S. Rodriguez, and Y. Shen. "Superradiant suppression of non-minimally coupled scalar fields for a rotating charged dS black hole in conformal Weyl gravity". *Class. Quant. Grav.* 43, 155026 (2026). arXiv: 2604.04352 [hep-th].

[2] N. Bao, J. Chu, Y. Jiang, and **J. March**. "Coarse-grained fixed-point tensor networks and holographic reflected entropy in 3D gravity". *JHEP* 02, 107 (2026). arXiv: 2509.10170 [hep-th].

Preprints

[3] N. Bao and **J. March**. "The Capacity of Entanglement and Holographic Entropies at Finite Resources" (Aug. 2026). arXiv: 2608.26377 [hep-th].

[4] N. Bao, K. Furuya, and **J. March**. "Combinatorial aspects of holographic quantum secret sharing" (July 2026). arXiv: 2607.16110 [hep-th].

[5] N. Bao, K. Furuya, and **J. March**. "Phase transitions and uberholography of holographic pure-state geometries" (July 2026). arXiv: 2607.06870 [hep-th].

Research

Quantum Information and Quantum Gravity

Northeastern University

- Investigating dualities between algebraic boundary CFT quantities and geometric bulk AdS quantities to deepen understanding of quantum gravity and quantum information.
- Developed a CFT-only computational framework that avoids holographic assumptions.

- Co-author on four papers spanning holographic reflected entropy from coarse-grained fixed-point tensor networks, combinatorics of holographic quantum secret sharing, uberholography and phase transitions of pure-state geometries, and the capacity of entanglement at finite resources (see Publications).

Quantum Error Correction and SYK Spectral Statistics

Northeastern University

- **Topological color codes.** Training a graph neural network decoder for the triangular color code, using the LightsOut formulation of color-code syndrome decoding to structure the learning problem.
- **SYK spectral statistics.** Learning the Sachdev–Ye–Kitaev spectral distribution with normalizing flows, in order to extrapolate beyond the reach of exact diagonalization to large N and probe deviations from random matrix theory.
- **Non-abelian codes.** Developing machine learning techniques for decoding in non-abelian topological codes.

Quantum Field Theory in Black Hole Backgrounds

Grinnell College

- Derived 2D holographic representations of 4D Kerr and Kerr–Newman–AdS black holes using scalar field probes.
- Applied finite element analysis to compute eigenmodes of non-spherically symmetric bound states.
- Demonstrated eigenspectral cancellation of non-S-wave modes with custom lightweight Mathematica code; co-author on the resulting paper in *Classical and Quantum Gravity*.

Hadron Jet Substructure Research, CMS Heavy-Ion Collaboration

Laboratoire Leprince-Ringuet, École Polytechnique

- Established the effectiveness of secondary Lund jet planes for isolating gluon-rich jet samples, using Monte Carlo event generation in C++/ROOT/Rivet.
- Contributed analysis to the study published as "Secondary Lund jet plane as a gluon-enriched sample," *JHEP* (2024).

Technical Skills

Computational Physics: Markov-chain Monte Carlo (C++ / ROOT / Rivet); finite element analysis (Mathematica); numerical methods; high-performance computing

Machine Learning: Python (PyTorch); graph neural networks and normalizing flows

Other: LaTeX; full-stack TypeScript; Linux, macOS

Teaching & Leadership

Teaching Assistant, Northeastern University, Dept. of Physics Sep 2024 – Present

- *Introductory Physics Laboratory* — delivered mini-lectures and oversaw data collection across a range of experiments for life-science students.
- *Advanced Physics Laboratory* — guided advanced undergraduates in designing and analyzing professional-grade experiments with publication-quality writeups.

Founder & Lead Developer, Supreme Physics Mar 2020 – Present

- Built and maintain an educational software platform supporting online tutoring, scheduling and payment systems.

- Trained additional tutors to deliver effective physics instruction.

Physics Workshop Teaching Assistant, Grinnell College, Dept. of Physics Aug 2023 – May 2024

- Supported students in hands-on, experiment-based introductory physics courses emphasizing conceptual understanding and collaborative learning.

Awards & Honors

H. George Apostle Prize in Physics, Grinnell College 2024

- Awarded annually to one or two graduating physics majors demonstrating exceptional academic performance, engagement, and promise in the field.

Grinnell Founder's Scholarship 2019 – 2024

Dean's List, Grinnell College 2019 – 2024

California Mathematics Council Award for Exemplary Mathematics Achievement 2019

Professional Memberships

American Physical Society · Society of Physics Students · American Institute of Physics